

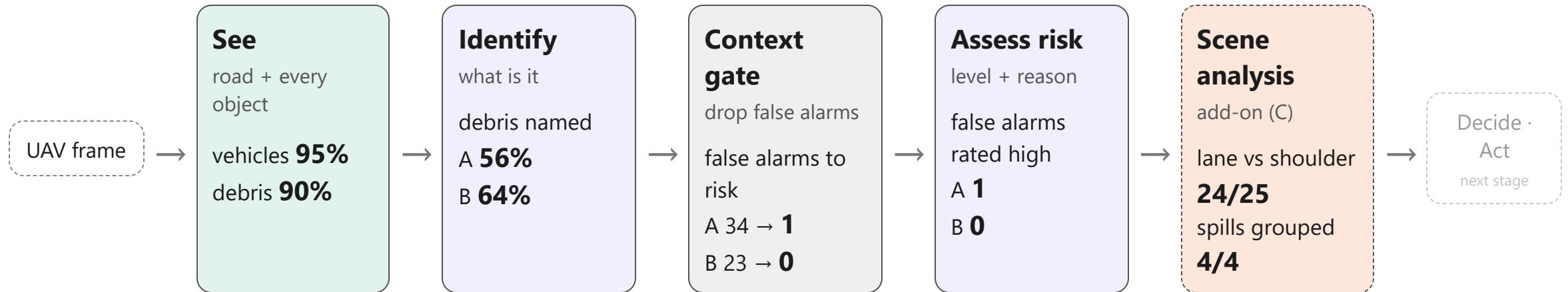
STAGE REPORT · 27 SEPTEMBER 2026

UAV-VisualSurv

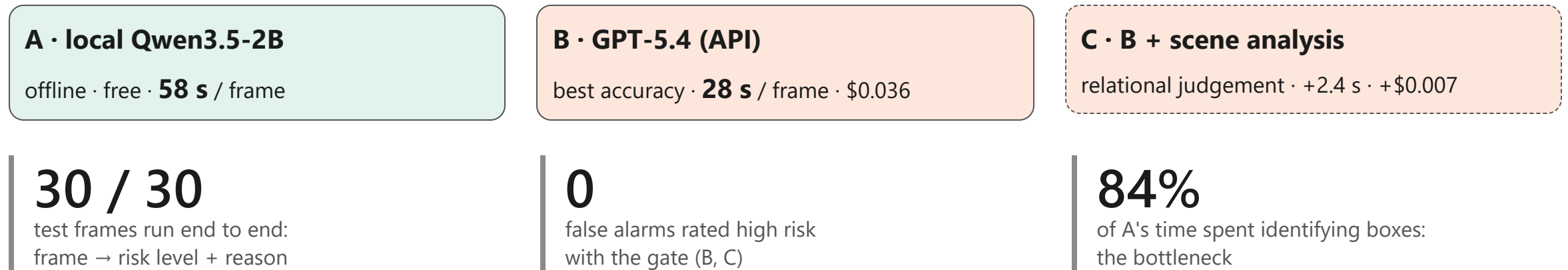
Highway hazard perception from a drone:
connecting the chain from seeing to risk, and comparing three architectures

This stage: connect the chain, compare the architectures

① Chain connected end to end · all 30 test frames



② Three architectures compared



Real UAV motorway frames + rendered 3D debris



Background

real UAV video, Pexels clip 8742752
Pexels License · open motorway only

Debris (red)

3D model "Low-Poly Treaded Tire"
by TepidGames · CC BY · Sketchfab
Blender 5.2.2 · random pose
real-world scale from calibrated
ground resolution (px per m)

Debris label

pixel-exact, from the render's alpha

Vehicle labels (blue)

Grounding DINO proposals
every box checked by hand

4 open-motorway locations

30 test frames · fixed

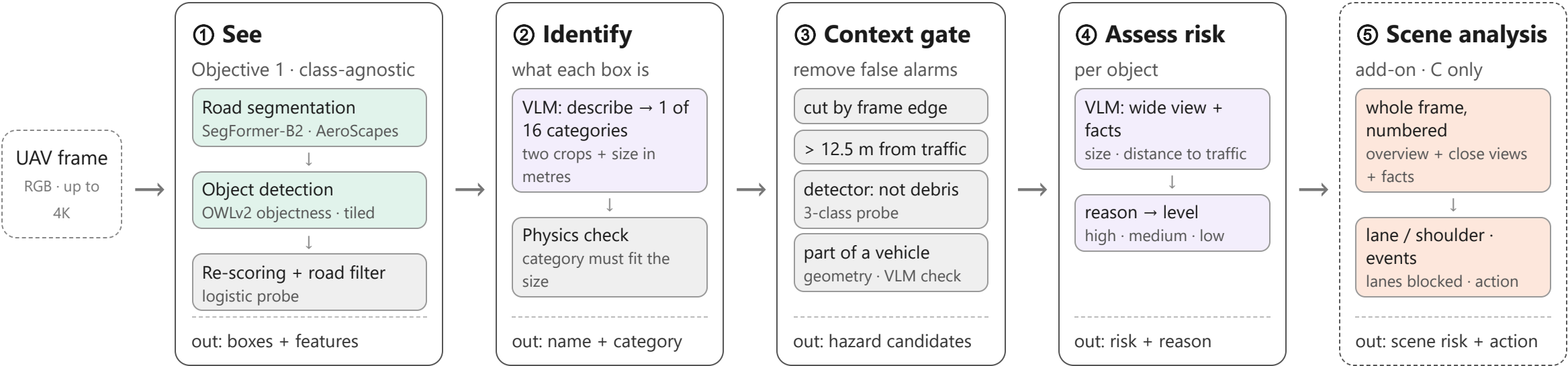
30 dev frames · all tuning

342 vehicles · hand-checked

1 debris object per test frame

leave-one-location-out scoring

One chain, three architectures



	① See	② Identify	③ Context gate	④ Assess risk	⑤ Scene analysis
A · local	local GPU	Qwen3.5-2B 4-bit · local	CPU (+ Qwen check)	Qwen3.5-2B · local	–
B · API	local GPU	GPT-5.4 · API	CPU (+ GPT check)	GPT-5.4 · API	–
C · B + add-on	local GPU	GPT-5.4 · API	CPU (+ GPT check)	GPT-5.4 · API	GPT-5.4 · API

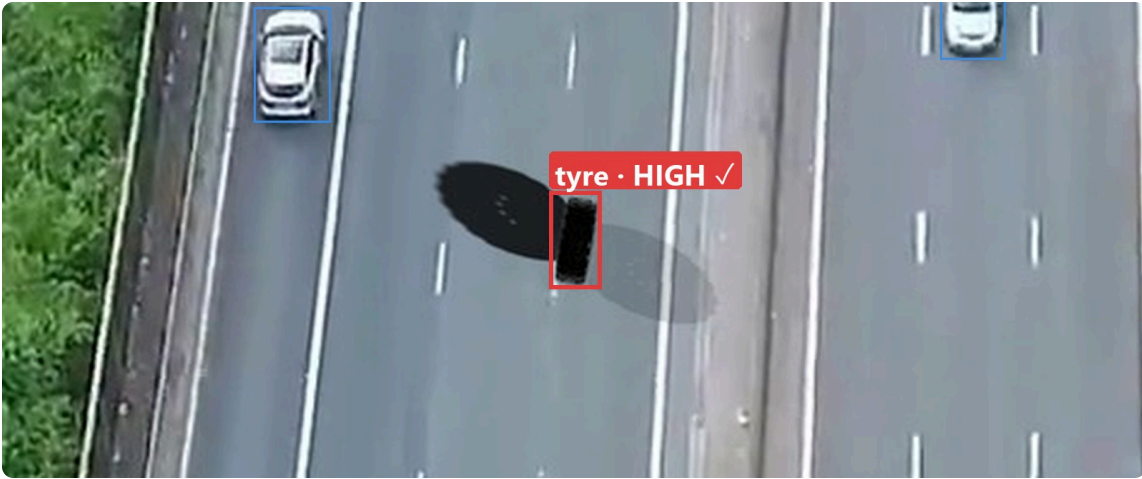
local GPU
CPU
cloud API
VLM, per architecture (table)

A · local Qwen3.5-2B

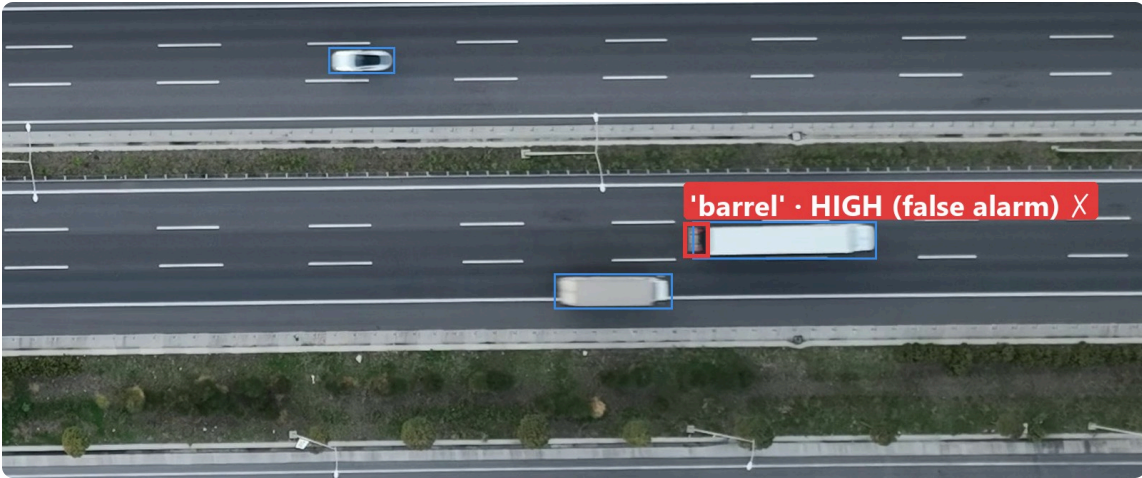
high medium removed by gate missed vehicle



✓ tyre found · HIGH · truck cut by frame edge removed



✓ small tyre found · HIGH



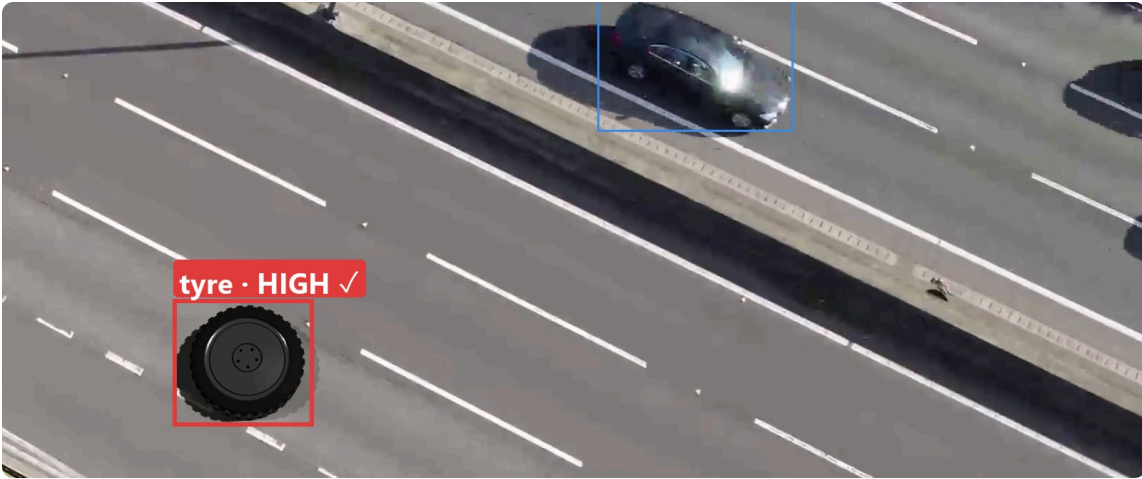
✗ lorry cab → 'barrel' · HIGH · false alarm reaches the alert



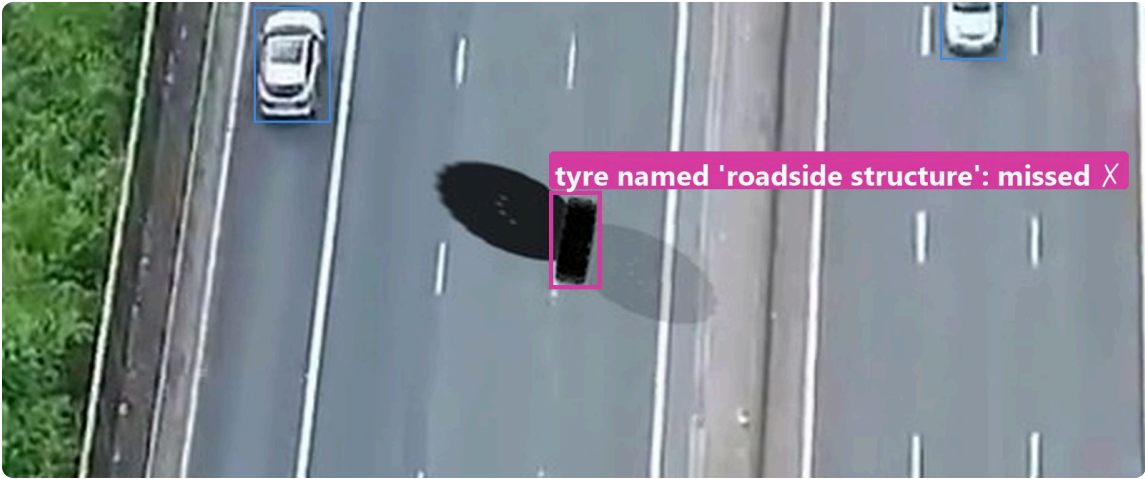
✗ fridge → 'roadside structure' · hazard missed

B · GPT-5.4

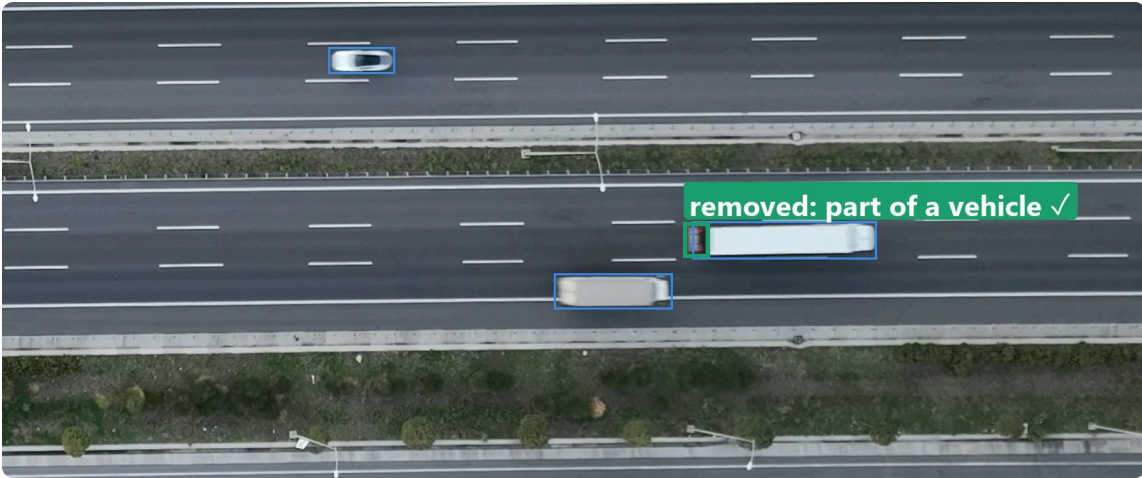
high medium removed by gate missed vehicle



✓ tyre found · HIGH



X small tyre → 'roadside structure' · missed



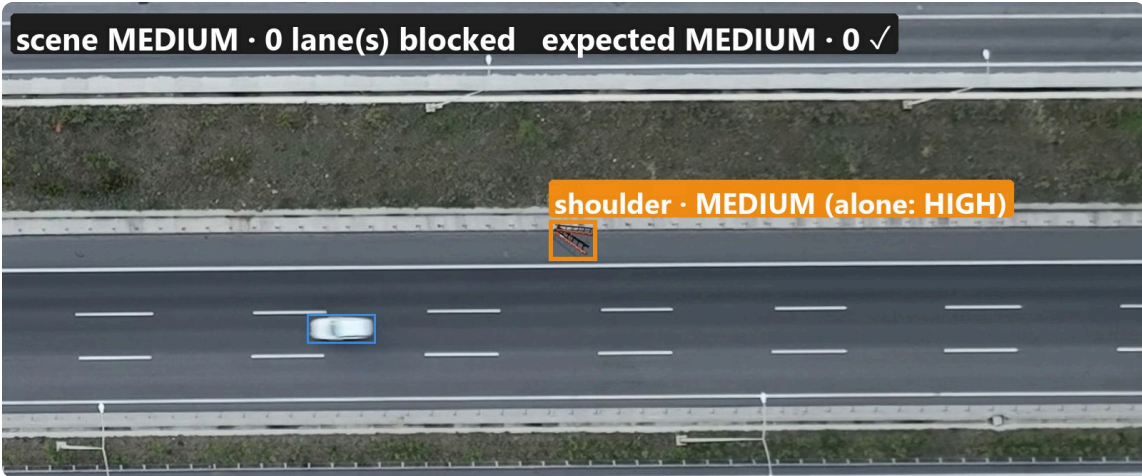
✓ lorry cab removed · GPT: "part of the vehicle"



✓ fridge found · HIGH · but it lies on the shoulder

C · scene analysis: the frame judged as a whole

23 scene-relation frames · lane / shoulder / spill / blockage / clear · answers fixed before running



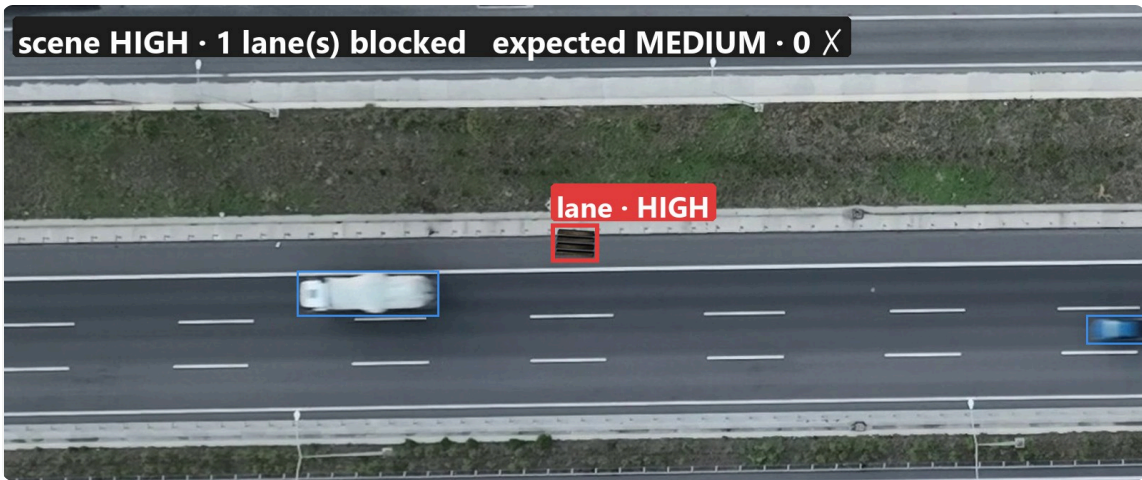
✓ shoulder object: HIGH alone → MEDIUM · "shoulder clearance"



✓ tyre + fridge side by side → 2 lanes blocked



✓ 4 planks → one spill from lorry V9 · 2 lanes · "close lanes"



✗ planks on the shoulder judged "in lane" · HIGH, expected MEDIUM

Time, accuracy and GPU in one table

	A · local Qwen	B · GPT-5.4	C · B + scene add-on
Inference time · s per frame · 30 test frames · RTX 3050 Ti (4 GB)			
road segmentation	0.32	0.32	0.32
object detection (OWLv2)	4.33	4.33	4.33
re-scoring	< 0.01	< 0.01	< 0.01
identification (VLM, per box) · bottleneck	49.2	22.0	22.0
context gate	0.09	0.13	0.13
risk assessment (VLM)	4.45	≈ 1.3	≈ 1.3
scene analysis (add-on)	–	–	2.4
total	58.4	≈ 28	≈ 30.5
Accuracy · 30 test frames			
detection: vehicles · debris · precision	95.2% · 90.0% · 84.1% (shared)		
debris named correctly	56%	64%	64%
vehicles named correctly	96.0%	96.4%	96.4%
false alarms removed by the gate · real debris lost	33 · 0	24 · 0	24 · 0
false alarms rated high	1	0	0
Accuracy · 23 scene-relation frames (add-on test)			
scene risk correct	16 / 23	16 / 23	17 / 23
lane vs shoulder · spill grouped · lanes blocked	–	–	24/25 · 4/4 · 16/23
GPU memory · peak			
Objective 1 pass · VLM pass	2.19 GB · 1.97 GB	2.19 GB · cloud	2.19 GB · cloud
all models loaded at once	4.05 GB > 4 GB card	2.19 GB	2.19 GB
Deployment			
API cost per frame · runs offline	\$0 · yes	\$0.036 · no	\$0.043 · no

Where we stand, what comes next

This stage

- ✓ chain connected: frame → objects → identity → gate → risk + reason
- ✓ "no false alarm rated high": met by B and C on the test set
- B: best accuracy and speed today
- A: offline and free, but slow
- C: add-on for relational decisions

Next stage

- decide + act: lane signals, message signs
- bottleneck: identify only unclear boxes, in parallel
- small-object detection (pallets ≈ 22 px)
- object size from altitude + focal length